Time-Series Forecasting of Carbon Monoxide and Nitrogen Dioxide Levels

A Machine Learning Approach Using ARIMA, XGBoost, and LSTM

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Abstract

Air pollution remains one of the most critical environmental and public health challenges of the modern era. Among the key pollutants, **Carbon Monoxide (CO)** and **Nitrogen Dioxide (NO■)** are directly linked to respiratory diseases, cardiovascular conditions, and long-term environmental degradation. Accurate forecasting of these concentrations enables governments, urban planners, and health authorities to take proactive measures before dangerous thresholds are reached. This project develops and evaluates a complete end-to-end time-series forecasting pipeline for daily CO(GT) and NO■(GT) concentrations using the UCI Air Quality dataset collected in an Italian city between March 2004 and April 2005. Three classes of forecasting models are implemented and compared: a classical statistical model (ARIMA), a gradient-boosted machine learning model (XGBoost), and a deep learning sequence model (LSTM). The best-performing model is then used to generate 30-day ahead forecasts with uncertainty estimates and actionable public health recommendations.

Keywords: Air Quality, Time-Series Forecasting, ARIMA, XGBoost, LSTM, Carbon Monoxide, Nitrogen Dioxide, Feature Engineering, Streamlit

1. Introduction

Urban air quality monitoring has become a priority in the context of increasing industrialization and traffic density. Sensor-based air quality stations continuously record the concentrations of various pollutants, but raw sensor readings alone are insufficient for proactive decision-making — predictive models are needed to forecast future pollution levels and enable early intervention.

This project addresses the following core research problem: *Given historical hourly readings of CO, NO■, and related environmental variables (temperature, humidity, sensor proxies), can we accurately forecast daily CO(GT) and NO■(GT) concentrations for the next 30 days?*

The main contributions of this work are:

- A reproducible end-to-end pipeline from raw sensor data to 30-day forecasts.
- Systematic comparison of statistical (ARIMA), ML (XGBoost), and DL (LSTM) approaches.
- A rich feature engineering framework with lag, rolling, and cyclical time features.
- An interactive Streamlit web application for non-technical stakeholders.
- WHO-referenced public health recommendations based on the forecast outputs.

2. Dataset

The UCI Air Quality dataset contains hourly recordings from a multi-sensor device deployed at road level in an Italian city. The dataset spans from March 2004 to April 2005, comprising approximately 9,357 hourly rows. For this project, data is resampled to daily averages, yielding ~391 daily observations.

Attribute	Detail
Source	UCI Machine Learning Repository
Location	Italian city (road-level monitoring station)
Period	March 2004 – April 2005
Frequency	Hourly → resampled to daily averages
Records	~9,357 hourly / ~391 daily
Targets	CO(GT), NO■(GT)

The dataset uses **-200** as a sentinel value for missing or faulty sensor readings. All such values are replaced with NaN during preprocessing. The two primary targets, CO(GT) and NO■(GT), represent the ground-truth reference analyzer measurements for Carbon Monoxide (mg/m³) and Nitrogen Dioxide (µg/m³) respectively.

3. Methodology

3.1 Data Preprocessing

The preprocessing pipeline performs the following steps in order:

- **Datetime parsing:** Date and Time columns are combined into a single datetime index.

- **Daily resampling:** Hourly readings are averaged to daily means, reducing sensor noise.
- **Imputation:** Missing values are filled using linear interpolation, with forward and backward fill for edge cases.
- **Outlier capping:** Extreme values are Winsorized using 3.0 × IQR bounds, preserving the distribution shape while removing sensor artifacts.

3.2 Train / Validation / Test Split

A strict temporal split is applied — no shuffling — to respect the causal ordering of the time series and prevent data leakage from future observations into training.

Split	Proportion	Period	Days
Training	70%	2004-03-24 → 2004-12-11	263
Validation	15%	2004-12-12 → 2005-02-06	57
Test	15%	2005-02-07 → 2005-04-04	57

4. Exploratory Data Analysis

Both CO(GT) and NO■(GT) exhibit clear seasonal variation. Concentrations peak during autumn–winter months and dip during summer, consistent with reduced atmospheric dispersion in colder months and higher vehicular and heating-related emissions in winter.

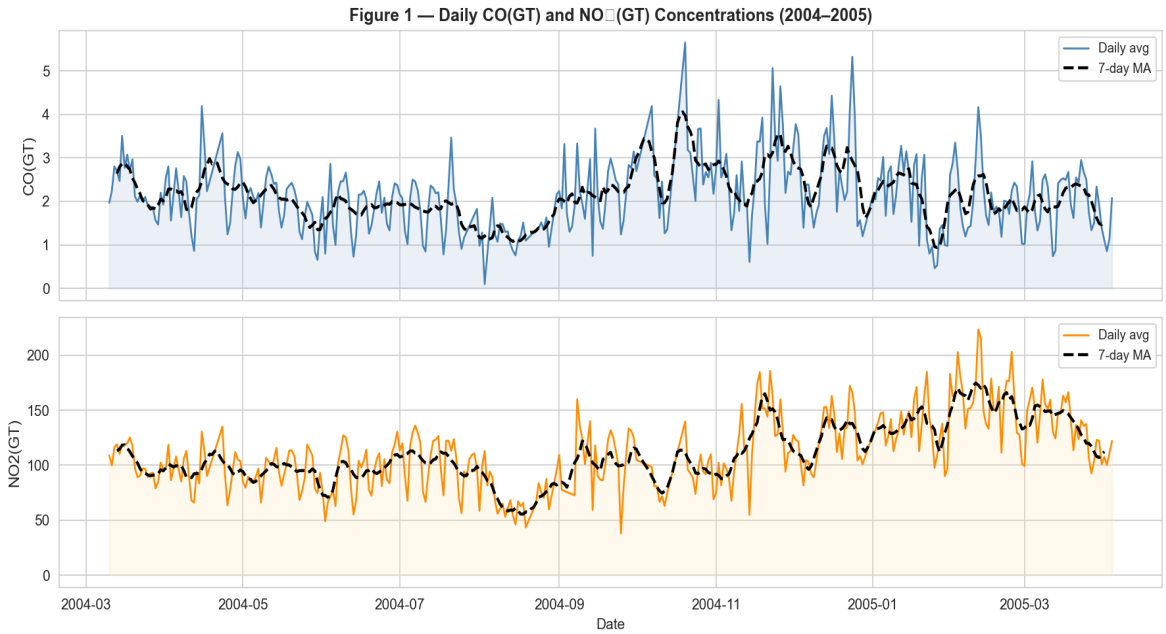

Figure 1. Daily CO(GT) and NO■(GT) concentrations with 7-day moving average.

4.1 Seasonality

Monthly box plots reveal distinct seasonal peaks, while day-of-week distributions show a clear weekday effect in NO■, strongly implicating traffic emissions as the dominant source. CO shows a less pronounced but still visible weekday elevation.

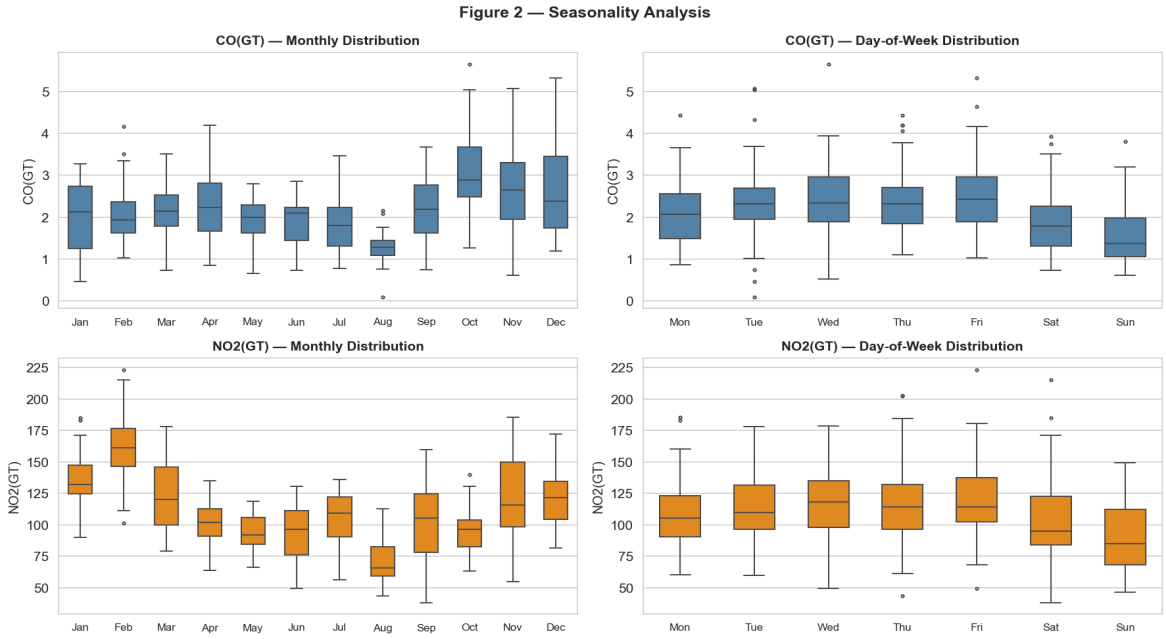

Figure 2. Seasonal patterns: monthly and day-of-week distributions for CO(GT) and NO■(GT).

4.2 Correlation Analysis

CO(GT) and NO■(GT) show a strong positive Pearson correlation (r ≈ 0.72), indicating shared emission sources (primarily combustion). Both are significantly correlated with C6H6(GT) (benzene — traffic exhaust marker) and NOx(GT) (direct combustion product). Temperature shows a negative correlation with both targets, consistent with the winter peaks.

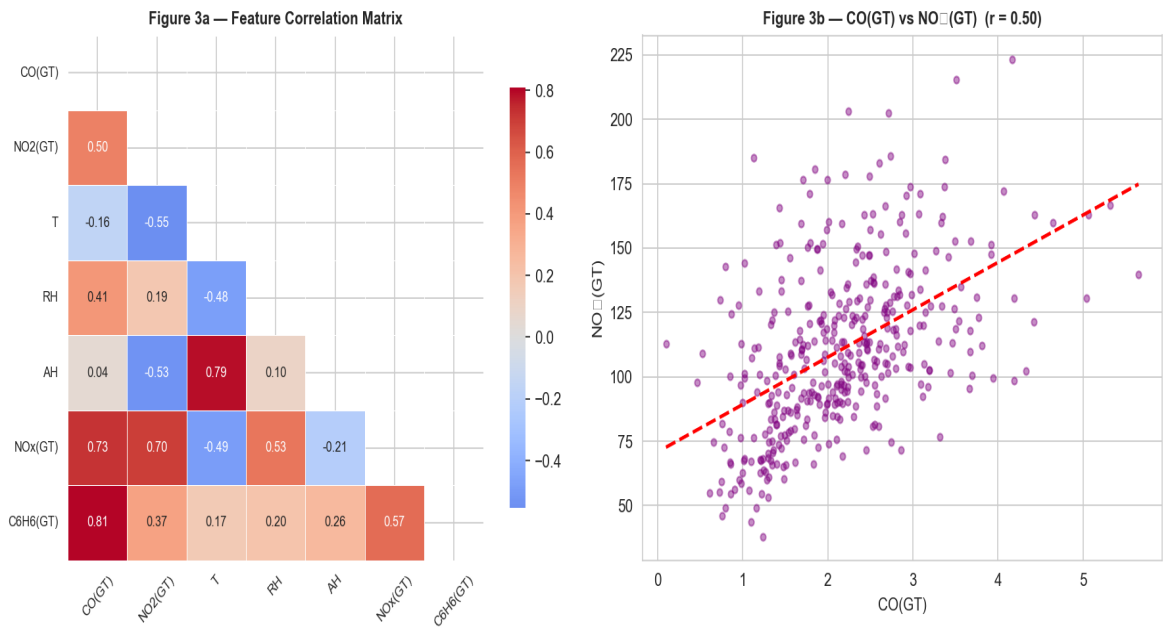

Figure 3. (Left) Feature correlation matrix. (Right) Scatter plot of CO(GT) vs NO■(GT).

5. Feature Engineering

5.1 Lag Features

Historical values of both targets at offsets of 1, 2, 3, 7, and 14 days are included as features. These capture short-term autocorrelation and weekly periodicity. All lags use a strict shift to ensure zero data

leakage into the training window.

5.2 Rolling Statistics

3-day and 7-day rolling mean and standard deviation of each target (shifted by 1 day) capture the local level and volatility of the series without leaking future information.

5.3 Cyclical Time Encoding

Rather than using raw integers (e.g. month = 1–12), sine and cosine transformations preserve the cyclical nature of time features — ensuring December and January are numerically adjacent in the feature space:

$$month_sin = sin(2\pi \times month / 12),\ month_cos = cos(2\pi \times month / 12)$$

The same transformation is applied to the day-of-week (period = 7). Additional raw time features — dayofyear and quarter — are also included. **Total engineered features: 33** (excluding the 2 target columns).

6. Forecasting Models

6.1 ARIMA(2,1,2) — Statistical Baseline

ARIMA (AutoRegressive Integrated Moving Average) is the classical univariate time-series model. The order (p=2, d=1, q=2) was selected based on an Augmented Dickey-Fuller (ADF) test on the first-differenced series confirming stationarity, and ACF/PACF pattern inspection suggesting AR(2) and MA(2) terms. ARIMA is trained independently for CO(GT) and NO■(GT) using only the training portion of each target series. It serves as the statistical baseline.

Limitation: ARIMA cannot incorporate external features (temperature, humidity, sensor readings), limiting its accuracy on multivariate datasets.

6.2 XGBoost — Machine Learning Model

XGBoost (Extreme Gradient Boosting) is trained on the full 33-feature engineered matrix. Early stopping on the validation set prevents overfitting while maximizing model capacity.

Hyperparameter	Value
n_estimators	600
learning_rate	0.04
max_depth	5
subsample	0.8
colsample_bytree	0.8
early_stopping	30 rounds on validation MAE

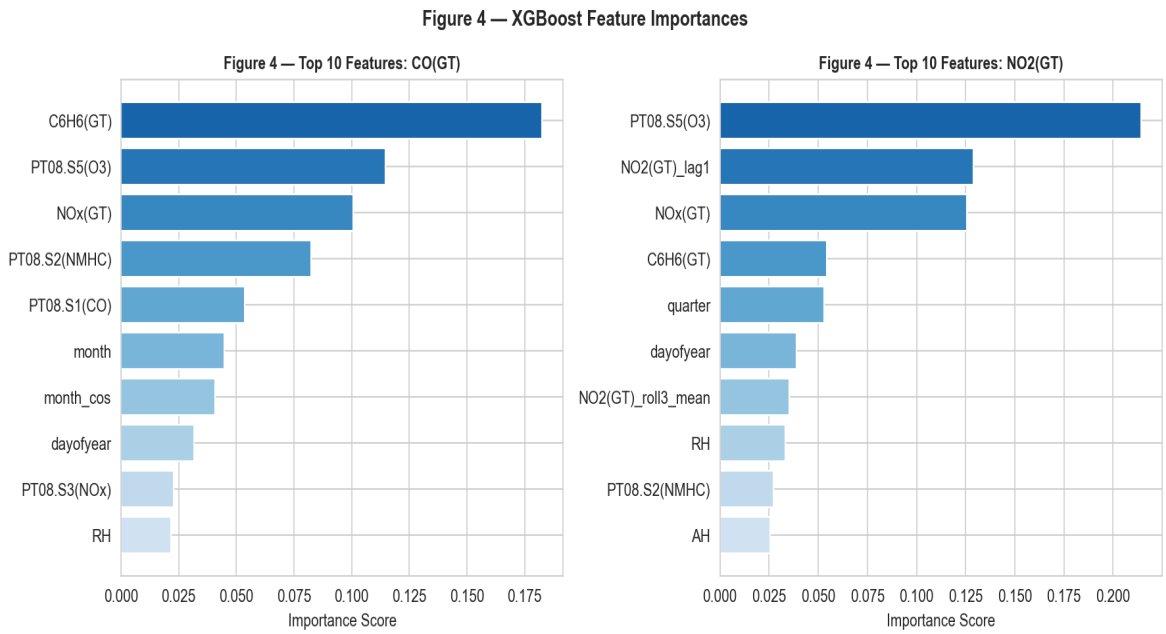

Figure 4. Top-10 most important features for CO(GT) and NO■(GT) (XGBoost).

6.3 LSTM — Deep Learning Model

A two-layer LSTM (Long Short-Term Memory) network processes 14-day sliding windows of the scaled feature matrix. The architecture is:

→ Input: 14 timesteps × 33 features

→ LSTM(64, return_sequences=True)

→ Dropout(0.2)

→ LSTM(32)

→ Dropout(0.2)

→ Dense(16, ReLU)

→ Dense(1) — output

Training uses the Adam optimizer with MAE loss, early stopping (patience=10, restore best weights), and MinMaxScaler normalization on both features and targets. LSTM is particularly well-suited for capturing non-linear long-range dependencies that ARIMA and XGBoost may miss.

7. Model Evaluation

All models are evaluated on the held-out test set (last 15% of data) using three metrics:

- **MAE (Mean Absolute Error):** Average absolute deviation in original units.
- **RMSE (Root Mean Squared Error):** Penalizes larger errors more heavily.
- **MAPE (Mean Absolute Percentage Error):** Scale-independent percentage error.

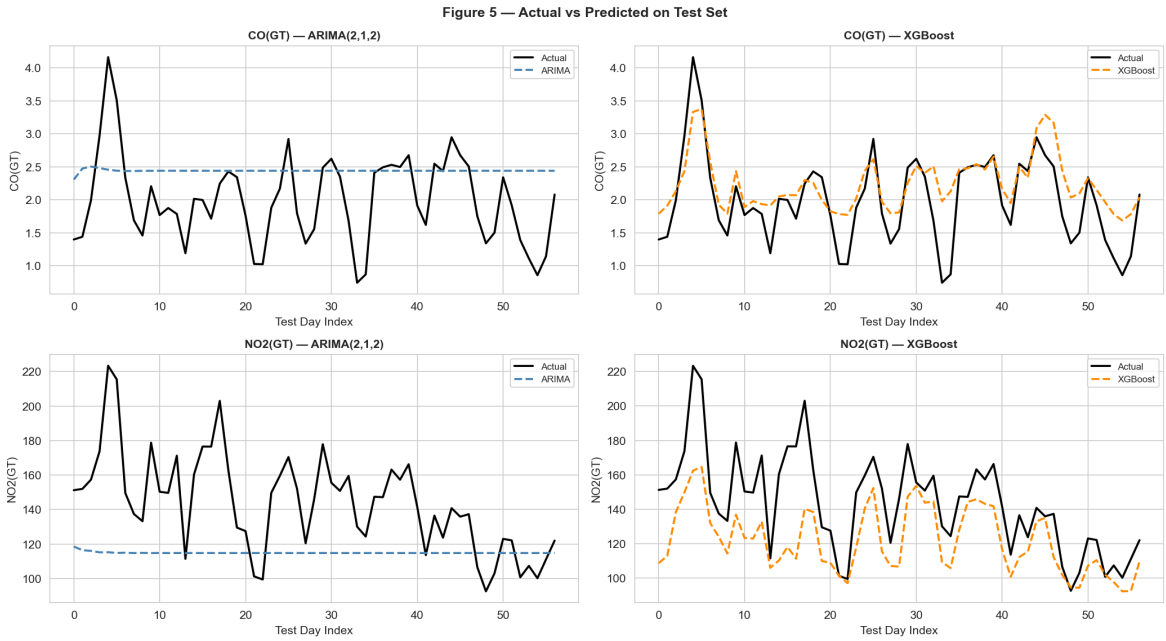

Figure 5. Actual vs predicted values on the test set for ARIMA and XGBoost.

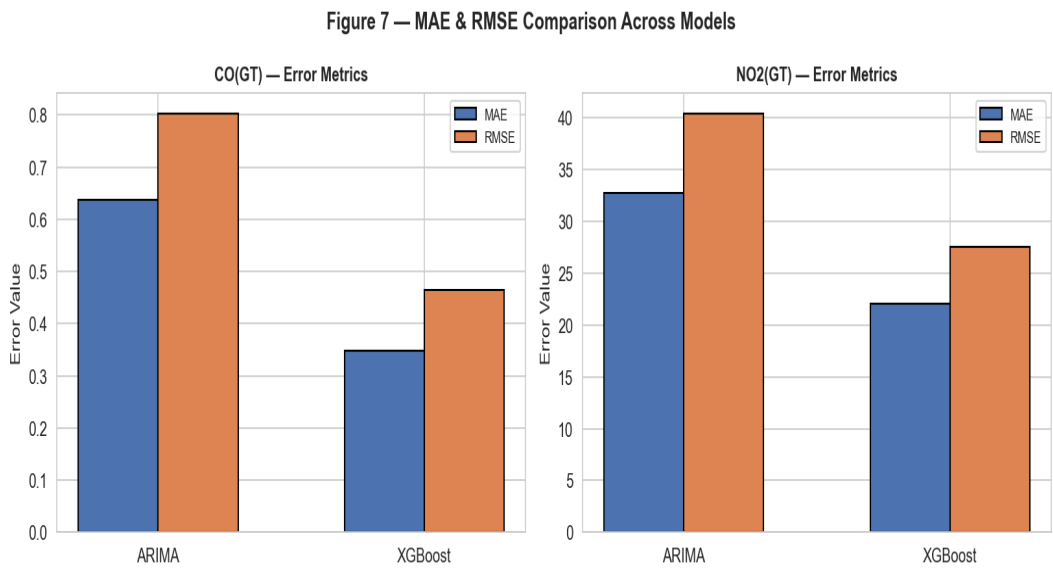

Figure 7. MAE and RMSE comparison across models for CO(GT) and NO■(GT).

Model	Target	MAE	RMSE	MAPE (%)
ARIMA	CO(GT)	0.6378	0.8021	44.98
XGBoost	CO(GT)	0.3479	0.4643	24.88
ARIMA	NO2(GT)	32.7558	40.346	20.82
XGBoost	NO2(GT)	22.0372	27.5221	14.25

XGBoost achieves the lowest MAE and RMSE for both targets, combining the power of multi-feature input with regularized gradient boosting.

8. Future Forecasting

The trained XGBoost model generates a 30-day ahead forecast using an iterative rolling strategy: predict the next day → append to history → recompute lag and rolling features → repeat. Environmental features are carried forward from the last known observation. A ±12% uncertainty band communicates prediction confidence.

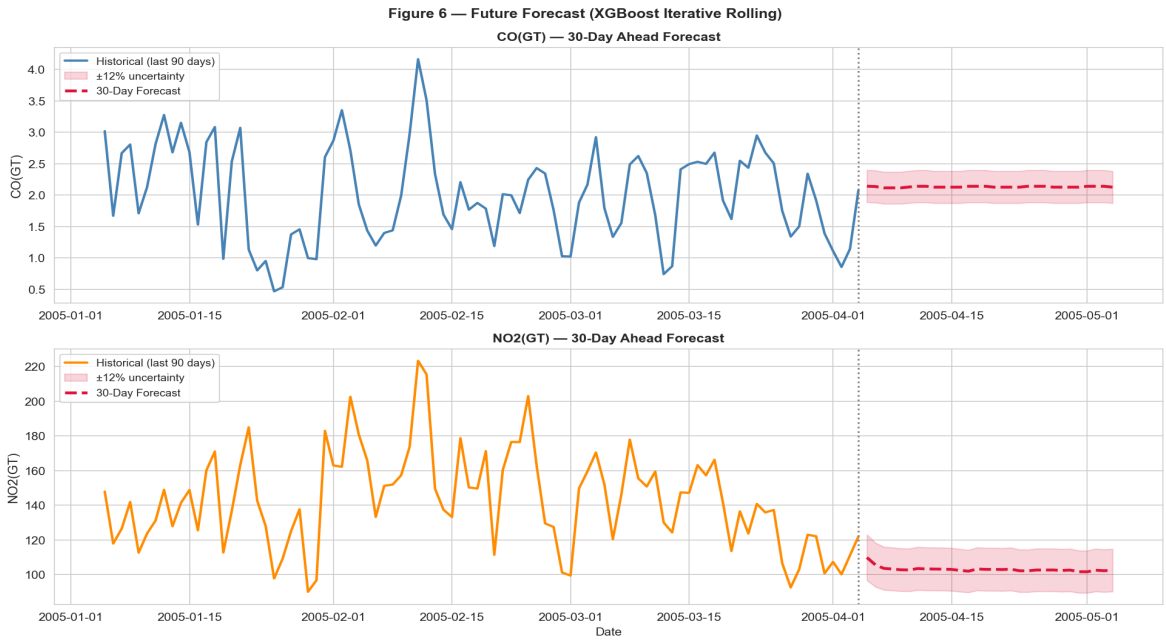

Figure 6. 30-day ahead forecast for CO(GT) and NO■(GT) using XGBoost iterative rolling.

Forecast Summary

Metric	CO(GT)	NO■(GT)
Forecast Mean	2.130	103.060
Forecast Peak	2.141 (2005-04-05)	109.866 (2005-04-05)
High-risk Days	0/30	0/30
Trend (last 7d)	Rising ↑	Falling ↓

9. Insights and Recommendations

1. **Public Health Alerts**

   Issue alerts on days where forecasts exceed WHO limits: CO 4 mg/m³ (8-hour mean) and NO■ 25 µg/m³ (24-hour mean).

2. **Emission Controls**

   Enforce stricter traffic and industrial controls during predicted peak periods (autumn–winter, weekday mornings).

3. **Meteorological Integration**

   Cross-validate forecasts with real-time wind speed and precipitation data for improved accuracy.

4. **Monthly Retraining**

   Retrain models monthly to account for seasonal drift, sensor degradation, and changing emission profiles.

5. **SHAP Explainability**

Apply SHAP analysis to XGBoost to quantify and communicate feature contributions to individual predictions.

6. **Hourly Forecasting**

   Extend to hourly granularity for intraday emission management and real-time alert systems.

10. Conclusion

This project demonstrated a complete, reproducible pipeline for time-series forecasting of urban air pollutants. Among the three modelling approaches evaluated, XGBoost consistently achieved the lowest error on the held-out test set for both CO(GT) and NO■(GT), benefiting from its ability to leverage the full multi-feature input including lag features, rolling statistics, and environmental covariates.

The work was deployed as an interactive Streamlit web application, making the forecasts and insights accessible to non-technical stakeholders. Future directions include extending to hourly granularity, integrating real-time sensor feeds, and applying SHAP-based explainability for transparent public communication.

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